THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL 1 EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

10 December 2014 TIME: 9:30 – 12:30 (3 hours)

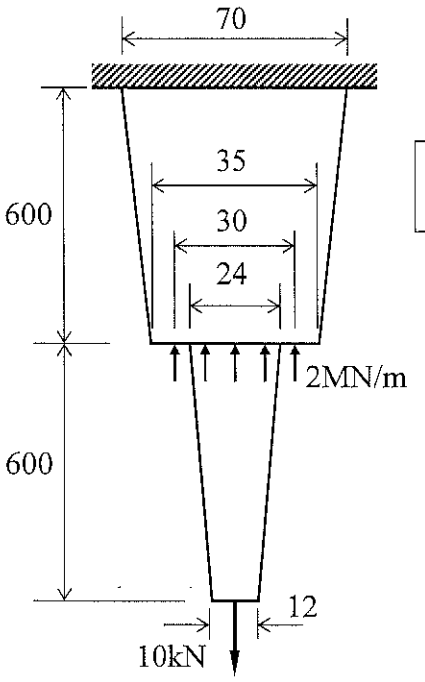
Answer **5 (FIVE)** questions.
All questions carry equal marks.
Marks for sub-questions are given.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A

1. (a) Two tapered circular bars, each with length 600mm, are rigidly connected along the common centre as shown in Figure Q1a. One of the big bar ends has fixed support against a wall. The diameters of the big bars are 70 and 35mm, while those of the small bar are 24 and 12mm.
A tensile point load of 10kN is applied at the free end at the centre of circle, and at the step change in section a compressive load of 2MN/m evenly distributed around a circle of 30mm diameter is applied. Youngs modulus of both bars is $E = 100$GPa.
Calculate the overall change in length of the tapered circular bars shown in FigureQ1a. **(12 marks)**



All units in mm, unless noted otherwise.

Figure Q1a

1. (b) Sketch the bending moment diagram (BMD) and shear force diagram (SFD) for the beam shown in Figure Q1b for $L > a$. The beam is subjected to uniformly distributed load. How can uplift happen at support A? Work out the relationship between a and L. How will this affect the BMD within AB? Sketch the resulting BMD. **(8 marks)**

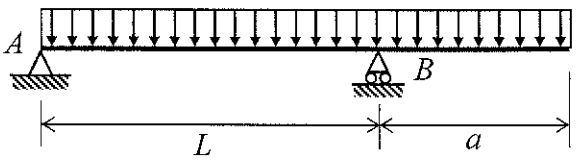
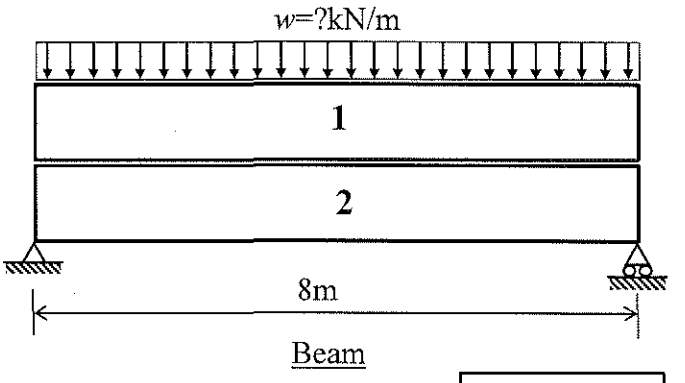


Figure Q1b

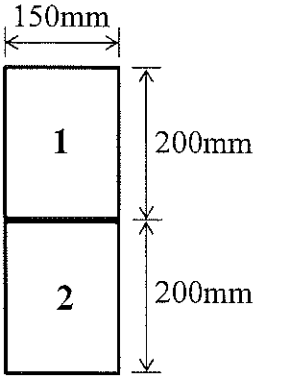
2. A simply supported beam having span of 8m is subjected to uniformly distributed load (UDL). The beam consists of 2 different materials with identical cross-section dimensions, stacked as shown in Figure Q2. Properties of the materials are as follows:
Youngs modulus: Material 1: $E_1 = 200{,}000$MPa
Material 2: $E_2 = 50{,}000$MPa
Allowable stress in tension and compression: Material 1: $\sigma_1 = \pm$ 80MPa
Material 2: $\sigma_2 = \pm$ 40MPa

(a) Calculate the maximum UDL w in kN/m if there is no composite action between Material 1 and Material 2. **(8 marks)**

(b) Calculate the maximum UDL w in kN/m if there is composite action between Material 1 and Material 2. **(12 marks)**



Beam



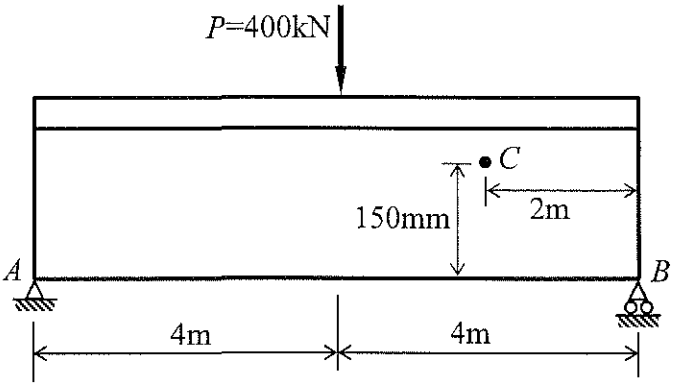
Cross section

Figure Q2

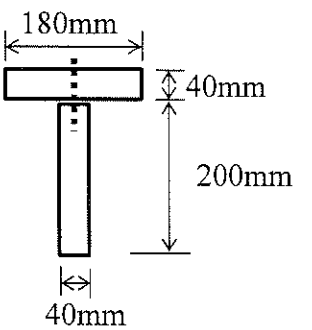
3. A simply supported beam AB with span of 8m is subjected to a point load $P = 400$kN at midspan, as illustrated in Figure Q3. The beam has a T-shaped cross section made of two steel plates (flange plate = 180×40mm, web plate = 200×40mm), connected by a series of bolts.

(a) Suggest the diameter and spacing of the bolts. The allowable shear stress of the bolt is 80MPa. **(10 marks)**

(b) Determine the magnitude of principal stresses and maximum shear stress at point C, where C is located 2m to the left of B and 150mm above the web base. **(10 marks)**



Beam



Cross section

Figure Q3